

Resistor Combination Simulator: Exploring Circuit Behavior Through Series, Parallel, & Complex Networks

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Abstract

This project presents the development of an interactive resistor combination simulator designed to demonstrate how different circuit arrangements affect electrical behavior. The simulation allows users to explore four circuit configurations: a single resistor circuit, a series circuit, a parallel circuit, and a complex series-parallel circuit. Users can adjust voltage and individual resistor values while observing changes in current, voltage distribution, and power throughout the circuit. The simulator includes visual circuit diagrams, animated current flow, and a physics insight section that displays the equations used for each circuit configuration. A component analysis feature provides detailed information about each resistor, including voltage, current, power, and dynamic observations explaining the role of individual components within the circuit. By combining mathematical calculations with visual and interactive representations, the simulation provides a comprehensive method for exploring how resistor combinations influence circuit behavior. This project demonstrates how interactive simulations can support physics learning by connecting theoretical relationships with observable circuit changes.

Introduction

Electric circuits are an important topic in physics because they describe how electrical energy is transferred and controlled through different components. A fundamental relationship used to analyze circuit behavior is Ohm's Law, which describes the relationship between voltage, current, and resistance through the equation $I = V/R$. While this relationship is straightforward for a single resistor, circuits containing multiple resistors require an understanding of how components interact within different arrangements. In series circuits, resistors share the same current while dividing the source voltage, whereas in parallel circuits, resistors share the same voltage while the current divides between branches. These relationships determine how changes to individual components affect the overall current, voltage distribution, and power of a circuit. Understanding resistor combinations is key for analyzing more complex electrical systems.

Several existing simulations demonstrate effective ways to teach resistor combinations in electrical circuits. Andrew Duffy's series and parallel circuit simulations allow users to adjust the source voltage and individual resistor values while observing how these changes affect voltage, current, and power throughout the circuit (Duffy, 2018). These simulations are effective because they provide immediate feedback and allow users to compare the behavior of different circuit configurations. However, they focus on individual circuit types rather than providing a unified environment for exploring multiple resistor arrangements. The simulation developed in this project builds upon these ideas by combining single resistor, series, parallel, and complex series-parallel circuits into one interactive application. In addition to real-time calculations, the simulation includes animated current flow, circuit-specific equations, and component analysis cards that provide detailed information about the behavior of each resistor, helping users connect circuit diagrams with the underlying electrical relationships.

Methods

The simulation was developed using HTML, CSS, and JavaScript in Visual Studio Code.

ChatGPT (GPT-5.5-mini, OpenAI) was used as an AI assistant throughout the development process. This process involved designing the user interface, programming circuit calculations, creating visual circuit diagrams, and implementing dynamic physics insights. The goal was to create an interactive circuit

simulation with various resistor combinations that allows users to adjust voltage and individual resistance values while observing how these changes affect current, voltage distribution, and power.

ChatGPT was used to assist with generating, debugging, and refining code, as well as providing guidance on interactive design features. The main prompts focused on creating a clean simulation that had selectable circuit types (single resistor, series, parallel, and complex series-parallel) with adjustable controls and animated current flow that followed the correct electrical paths. Additional prompts focused on implementing calculations for equivalent resistance, current, voltage drops, and power, along with developing component analysis cards that displayed resistor-specific values and observations. Examples of major prompts included:

- Design an interactive circuit simulator using HTML, CSS, and JavaScript with real-time sliders for voltage and resistance adjustments.
- Create selectable circuit diagrams for single, series, parallel, and complex resistor combinations using a dropdown menu.
- Animate current flow through each circuit while ensuring the animation followed the correct electrical path and stopped when voltage was set to zero.
- Calculate total circuit values and individual resistor values using appropriate series and parallel circuit equations.
- Develop dynamic physics explanations and component analysis sections that update based on circuit configuration and user adjustments.

AI-generated code was not implemented without first being reviewed. Each generated section was tested within the simulation, modified when necessary, and compared against expected physics relationships. The calculations were verified by confirming the resistor combination equations. Visual features, such as current animations and resistor highlighting, were tested by changing circuit types and adjusting the controls to ensure the simulation behaved correctly.

The development process required several iterations. Early versions focused on establishing the basic circuit calculations and user controls. Later iterations improved the circuit drawings, current animations, and component analysis cards. Several challenges occurred during development, including having diagrams that correctly lined up, having the animated current follow the correct path, preventing current flow when voltage was zero, and improving the layout to make the simulation easier to use. These issues were addressed through repeated testing, debugging, and refinement.

AI was especially helpful for quickly identifying programming errors and explaining JavaScript concepts during development. However, human judgment was still required to determine whether the simulation accurately represented physics concepts, decide which features improved the learning experience, and evaluate whether the generated code met the design goals of the project. All AI-generated code and written content used in the final submission was reviewed, tested, and verified before being included.

Results

The completed simulation is an interactive resistor combination simulator that allows users to explore how different circuit arrangements affect electrical behavior. Users can select between four circuit configurations: a single resistor circuit, a series circuit, a parallel circuit, and a complex circuit containing both series and parallel components. The simulation includes adjustable voltage and resistor controls that allow users to modify circuit parameters and observe how these changes affect current, voltage distribution, and power in real time. Each circuit configuration applies the appropriate resistor combination equations, which are displayed through the physics insight section to help users understand the calculations being used. This section of the simulation can be seen below in **Figure 1**.

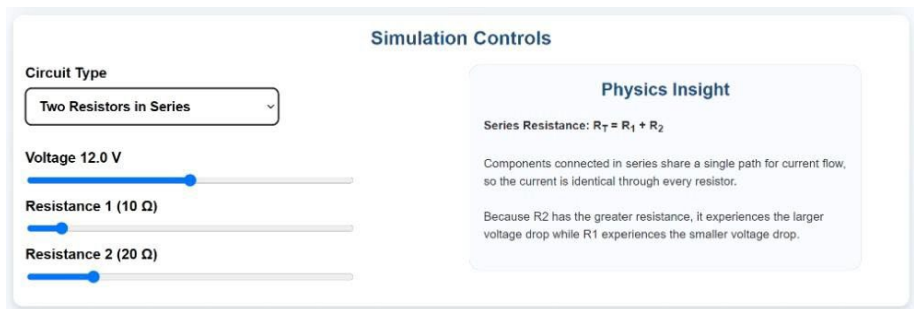


Figure 1. Circuit Selection & Insight

The simulation provides a visual representation of each circuit through labeled and color-coded circuit diagrams. Animated current flow is included to visually represent the movement of current through each circuit pathway and updates based on the selected circuit configuration. The circuit diagrams update in real time as users modify resistor values, allowing users to observe how changes to individual components affect the overall circuit behavior. The single resistor, series, and parallel configurations demonstrate different resistor arrangements, while the complex circuit combines these concepts to show how series and parallel relationships can exist within the same circuit network. This section of the simulation can be seen below in **Figure 2**.

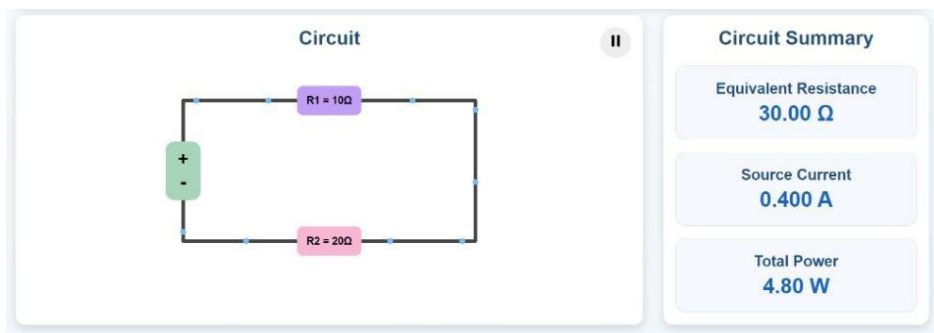


Figure 2. Circuit Diagram & Summary

A major feature of the simulation is the component analysis section, which provides detailed information about each individual resistor within the selected circuit. The analysis cards display the voltage, current, and power associated with each resistor, along with dynamic observations that explain the role of each component within the circuit. These observations update based on the selected circuit configuration and demonstrate concepts such as voltage division in series circuits, current division in parallel circuits, and component interactions in mixed series-parallel circuits. Each resistor is color-coded and linked to its corresponding location in the circuit diagram, allowing users to connect calculated values with the physical behavior of individual components. This feature provides a more detailed understanding of how each resistor contributes to the overall behavior of different circuit arrangements. This section of the simulation can be seen below in **Figure 3**.



Figure 3. Individual Resistor Analysis

Discussion

A strength of the simulation is its ability to combine visual and mathematical representations of resistor combinations. The color-coded circuit diagrams, dynamic physics insight as to what equations are relevant, animated current flow, and component analysis cards help connect calculations to the physical behavior of individual resistors within a circuit. A limitation of the current simulation is that users are limited to the provided circuit configurations rather than being able to create their own resistor networks. Additionally, the simulation currently focuses only on resistors and does not include other circuit components. Future improvements could include a custom circuit-building feature, additional components such as capacitors, and an interactive quiz or challenge mode. For example, users could be given a target current or power value and challenged to adjust voltage and resistance values to achieve the goal, encouraging active problem-solving and deeper understanding of circuit relationships.

Conclusion

This project resulted in the development of an interactive resistor combination simulator that demonstrates how different circuit arrangements affect electrical behavior. The simulation allows users to explore single resistor, series, parallel, and complex circuits while observing changes in current, voltage, and power through real-time calculations, animated circuit diagrams, and component-level analysis. By combining visual representations with numerical data and explanations, the simulator provides a more comprehensive way to understand how individual resistors contribute to the behavior of larger circuits. Future improvements could expand the simulator by allowing users to create custom circuits, adding additional electrical components, and incorporating interactive challenges or quizzes. These additions would further encourage exploration and problem-solving while extending the simulation into a more complete learning tool for understanding circuit concepts.

References

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